

Propeller. Design of the propeller plays an important role in smooth propulsion and ensuring long flight times. A propeller can spin in either clockwise or anti-clockwise direction. Two- or three-bladed propellers are commonly seen in drones.

- Select propeller size based on the chosen motor to prevent overheating. A propeller is mounted on the motor so that the two can spin in sync with each other. Check that the propeller comes with adaptor rings as a backup option in case the propeller does not fit the motor.

- For a stable flying experience, large blades are ideal. But if you plan to change speed frequently, smaller blades are better.

- For beginners, chances of crashing are high, so using plastic is a good option due to its low cost. Just like for frames, if you want durability and high quality, go for carbon-fibre instead.

Wood is another alternative that can be utilised for toughness.

- Use prop saver and prop guard to keep the propeller in position and prevent it from coming in contact with the external environment in case of a light crash. While adding these, keep in mind overall weight.

Battery. Like most electronic devices nowadays, drones use high-cost lithium-polymer batteries as these are lightweight and provide high power.

- Use a battery with higher capacity (amp-hours) for more flight time. But remember, higher capacity means more weight. While starting, maximum flight time of fifteen to twenty minutes is suitable.

- The battery's voltage must match that of the motor(s).

- Avoid safety issues. If the battery has not been charged properly, or it has been used for long enough for it to get discharged mid-air, it might burn. Also, there is a high probability of burning during a crash. Use a battery safety sack for prevention. You can use a battery beeper to inform when the battery is about to get discharged. Many commercial drones have a battery monitor to keep a tab on flying time and charge levels.

Electronic speed controller (ESC). An ESC connects to the flight control-

ler, motor and battery, and aids in controlling speed and direction of the motor(s).

- A power distribution board is used to share main battery power with different parts, and terminals so provided must be easy to solder to.

- The ESC provides the right voltage and handles maximum current consumption. Design the battery elimination circuit to convert DC voltage of the battery to DC voltage of the flight controller.

- Additional firmware can make the ESC more responsive to input changes.

Flight controller. This is the brain of the drone as it consists of microprocessors, sensors and I/O pins necessary to operate a UAV in air. Different sensors are used to obtain such information as height (barometer), speed, direction (compass), etc.

- Keep the flight controller near the horizontal centre of the drone to maintain the centre of gravity.

- Commonly-used sensors, including accelerometer (linear acceleration) and gyroscope (angular acceleration), help in providing stability while flying. These are contained in the inertial measurement unit (IMU) of commercial drones. With an increase in the number of sensors, cost increases, too.

- GPS helps in determining drone location at a specific point of time using satellites. Even when the drone flies away and is not visible, you can set the coordinates to which it should go for flying accurately. Set a home point so that the drone can return safely even if connection is lost.

Building the drone

After getting all components, start building the UAV.

- Improve your soldering skills if you are a beginner to ensure neat connections, higher reliability and less resistance. Ensure correct polarity while soldering.

- Insulate the electrical connections with insulation tubes to prevent short-circuits. Use heat to shrink the tube and, hence, insulate the components.

- Use hot glue for mounting components. Apply thread locking for connections that use bolts and nuts. This

technique keeps connections together but, at the same time, components can be removed when needed.

- You can add LEDs to the drone to increase visibility at night.

- Use plastic, metal or rubber materials for landing to avoid any damage to components, especially for drones with payloads.

Flying

While flying, the transmitter or remote controller is used for controlling the UAV and standard receiver unit to catch signals. A good antenna can work at a larger distance. There are many options available in the market. Most remote-controlled drones use 900MHz for transmission.

- Use simulators (like RC Drone Flight Simulator 3D) to learn basic flying before doing an actual one.

- Know your country's drone policy for operation by civilians and companies.

- First, fly only a little high in the air and check if motors are working. Never fly with a damaged or bent propeller. Inspect different parts and keep spare ones for backup.

- Take permission from nearby people before you fly. Arrange for a large open space where nothing could get in the way of flying the drone.

- Also, unless components are waterproof, do not take the drone out in rain, snow, etc, as it will damage the components. You can attach cameras (like, GoPro) if you wish to record the view in the final model.

- While self-levelling mode is easier for beginners, you should fly in acro or full manual mode to have complete control over the drone. You can use self-levelling as an alternate mode to avoid crashing.

Some cool features

Most ESCs allow propellers to spin only in a single direction. If you wish to rotate the propellers in both clockwise (CW) and anticlockwise (CCW) directions, use two different sets of propellers. You can also utilise firmware to configure these to fly in two directions.

You can also make the drone waterproof by using materials like carbon-fibre for arms and plastic for electrical parts. **EFY**